Matthew Cho

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EDUCATION

McGill University (GPA: 4.0/4.0) | M. Sc. in Biological and Biomedical Engineering 2023 – December 2025
McGill University (GPA: 3.8/4.0) | B.Eng. in Bioengineering, graduated with distinction. 2018 – May 2022

SKILLS

Programming: Python (Pandas, Scikit-Learn, PyTorch), Java, MATLAB, SQL, JavaScript, HTML/CSS, C, Git.
Analytical: Statistical analysis, data manipulation & visualization, data structures & algorithms, machine learning.
Laboratory: Cell culture, PCR, electrophoresis, optical microscopy, animal handling and husbandry, technical writing.
Academic/other: Fluently bilingual (French), research, critical thinking, tutoring, presenting, teamwork, leadership.

PUBLICATIONS

Cho, M. (2026). *ProMS: An Integrated Computational Framework for Molecular Surface Analysis and Atomic-Level Hydrophobicity Mapping* (Master's thesis). McGill University.

Perumal, F. A. S., **Cho**, **M.**, et al. (2026). Amplification of computational power by the multiplication of bacteria exploring microfluidic networks encoding mathematical problems. *Nature Microsystems & Nanoengineering*.

Cho, M., et al. (2024). Protein adsorption on solid surfaces: Data mining, database, molecular surface-derived properties, and semiempirical relationships. *ACS Applied Materials & Interfaces*, 16(22), 28290–28306.

EXPERIENCE

Research Assistant | McGill University May 2021 – December 2022, December 2025 - Current
- Investigated molecular surfaces and their properties under the supervision of Professor Dan Nicolau.
- Analyzed, designed, and deployed novel computational models of biological agent behaviour (bacteria and fungi).
- Built a website (HTML, CSS, JavaScript) connected to a MySQL database to display protein adsorption data online.
- Served as Teaching Assistant for a graduate-level course and supervised several undergraduate research students.

Summer Intern in Toxicology | Charles River Laboratories May 2019 – July 2019
- Performed in-life technical procedures on four animal species (rats, mice, rabbits, and dogs).
- Measured bodyweights and food consumption, maintained excellent animal welfare, held for dosing and blood tests.

PROJECTS

Protein Molecular Surface Calculator (ProMS) | Python, C
- Developed an integrated, object-oriented Python program with a simple GUI to characterize molecular surfaces.
- Implemented custom algorithms to quantify and visualize the surface properties of proteins and small molecules.
- Deployed the application as an open-source extension (plugin) of Schrodinger's PyMOL.
- Distributed the tool to researchers and collaborators across multiple universities and biotechnology organizations.

Computational Simulations with Biological Agents Suite | Python, Java, C, MATLAB
- Designed computational programs featuring custom algorithms to simulate the behaviour of biological agents.
- Developed an interactive visualization module with a graphical user interface for analyzing simulation outputs.

Tennis Match Predictive Model | Python
- Scraped online databases to collect thousands of data values. Cleaned and analyzed the data to discover trends.
- Built a logistic regression model which predicted the outcomes of upcoming tennis matches based on processed data.
- Model had a 67% prediction rate and a 6% ROI (compared to betting markets) over a sample size of 2000+ games.
- Programmed a second model to predict the result of baseball (MLB) games using sabermetric principles.

AWARDS

McGill Engineering Undergraduate Student Master's Award (MEUSMA) McGill University, 2023
J.W. McConnell Scholarship McGill University, 2018
Governor General's Academic Medal Governor General of Canada, 2017